

Draper Internship - review and advice! 🪐

Julia McGillivray

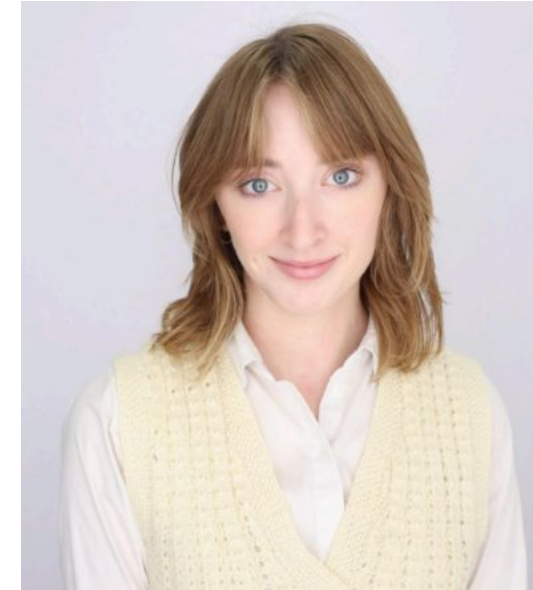


About Me



Julia McGillivray

- Senior studying Mechanical Engineering, Aerospace concentration
- Originally space concentration - now aero!
- From Boston, MA - go pats!
- Engineers without Borders , DBF
- Admissions Representative (aka tour guide :))
- A.O.E Secretary - Women in Engineering
- Mechanical Engineering Intern at Draper

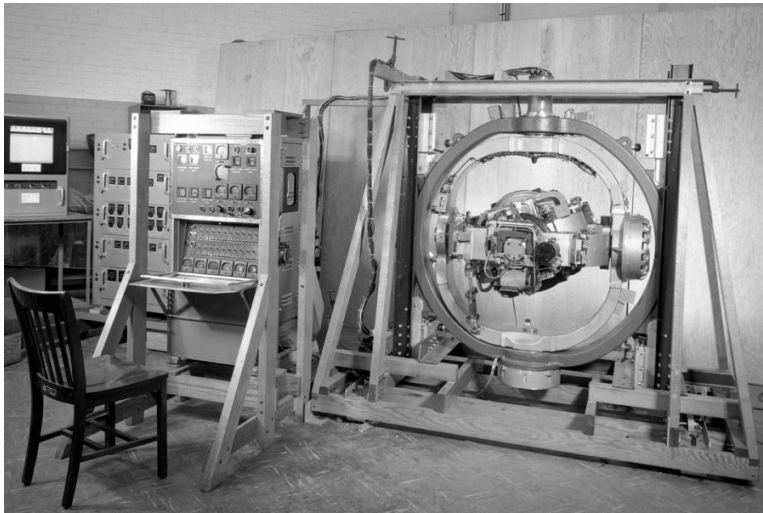
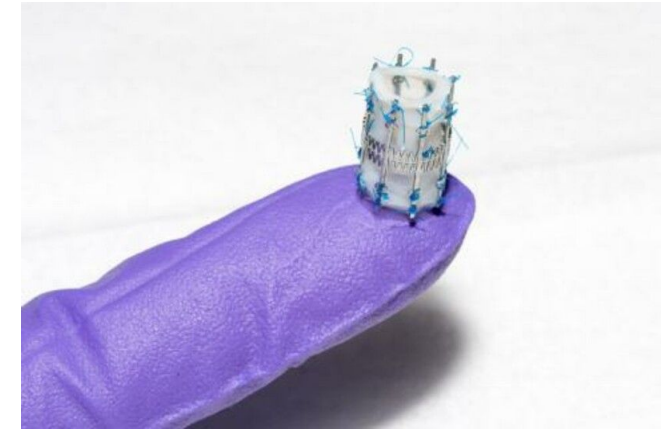




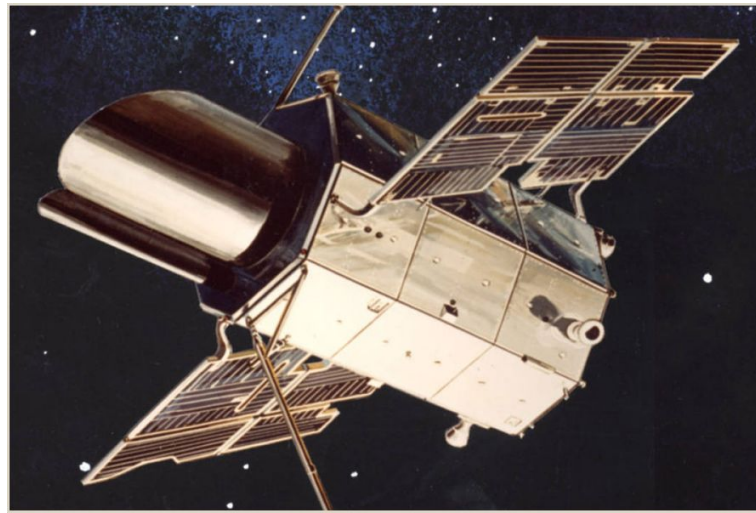
About Draper



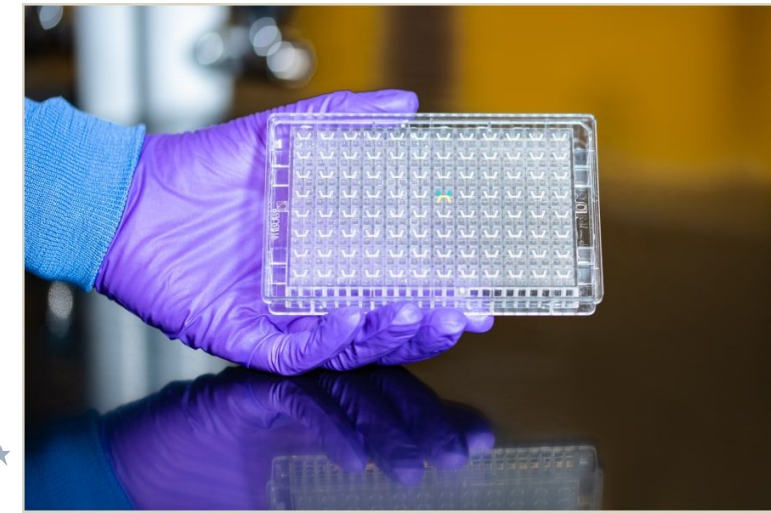
- An MIT Laboratory started in the 1930s from Charles Stark Draper, branched into its own company in the 1970s
- Notable projects: Apollo missions, invention of the IMU, heart valve projects with Boston Children's Hospital
- Contractor for cutting-edge technology with NASA, US Military, Lockheed Martin, etc.



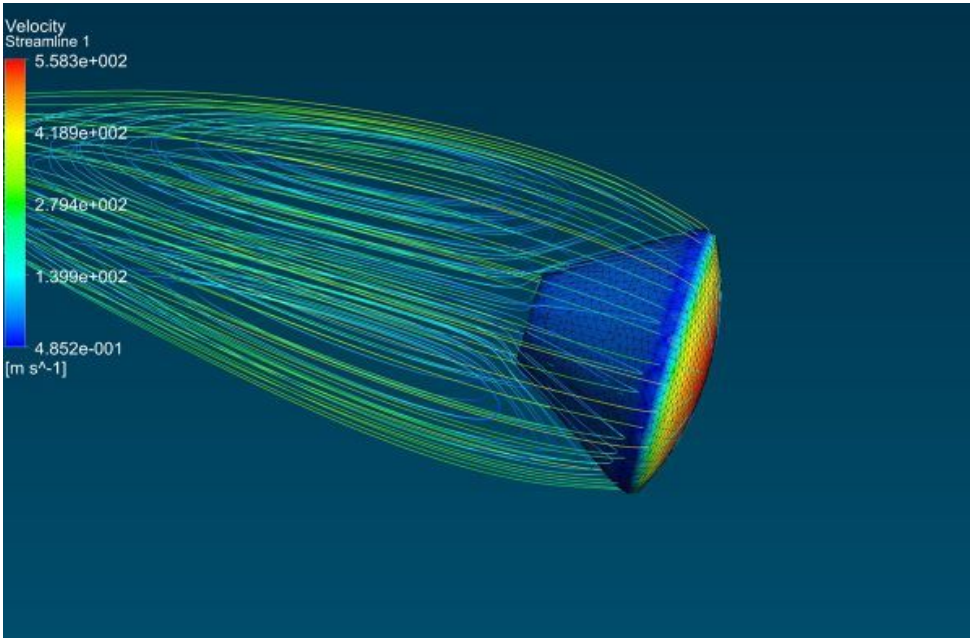
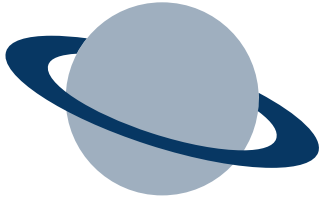
Inertial Measurement



NASA IMU



Organ on a chip

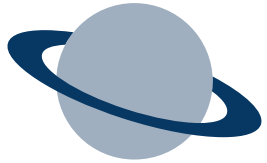


My job

- Aerodynamic efficiency and analysis of aerospace applications
- Multi-objective optimization - how to make something fly as efficiently as possible dependent on geometric characteristics!
- Computational Fluid Dynamics - what will this design act like in harsh environments?



Skills and softwares I used



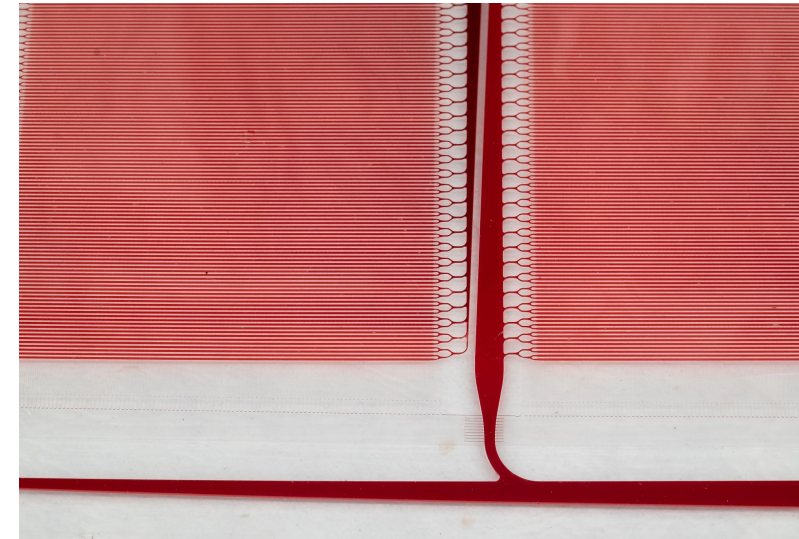
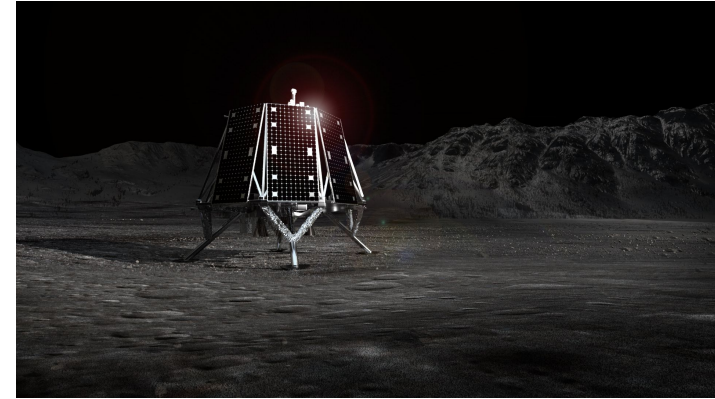
- A lot of Python. Pay attention in Engineering Computations!
- MATLAB - was able to integrate code into both.
- ANSYS - I used Fluent for flow simulation. Can either model things or simulate them.
- Others - depending on the field and company, they might have you learn softwares that are not open source or public access. Be prepared to pick those things up!
- Some companies also make their own platforms.
- Basic things like Powerpoint, Teams, etc.



Some tips and tricks



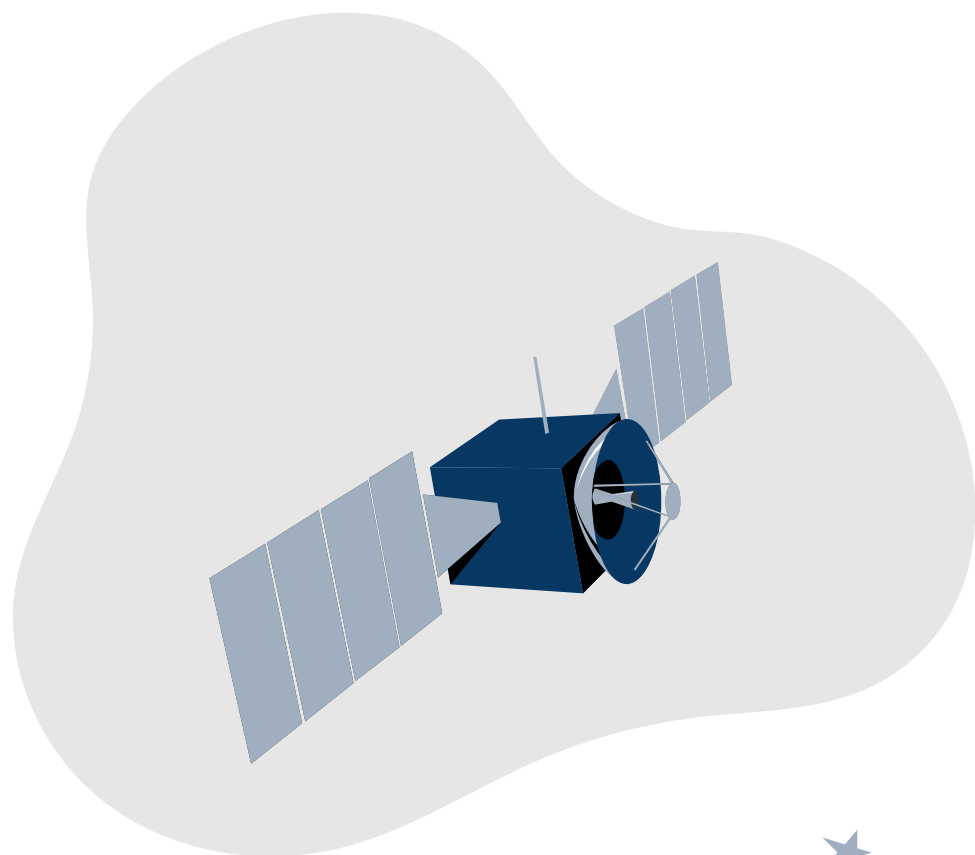
- ★
 - Don't be nervous about not knowing things. Interns never know things right away.
- ★
 - Be eager to learn and pick up new things. Draper cared a lot more about my interest and capability rather than prior knowledge.
- ★
 - They will look more at your potential and ability to work with people way more than doing everything perfect!
- ★
 - Cold DM alumni from GW on LinkedIn. Easiest way to get a connection at a company if you don't have one.





What I liked!

- Working with professional engineers will increase your skills tenfold.
- A full-time engineering environment was really rewarding, and I loved being shown the real project processes.
- Full-timers are often really chill. They want to welcome you and get to know you!
- Other interns are very fun to work with!



Questions or comments?